

Terrace Restoration and Management in Steep-Slope Viticulture

Spatial scale: vineyard / landscape

About this sustainable management strategy

Terrace restoration and the creation of new terracing systems are land management practices commonly used in steep mountain and hillside viticulture. Terrace systems consist of a series of level or gently sloping platforms supported by retaining walls or embankments, enabling grape production on terrain that would otherwise be difficult to cultivate. Traditional dry-stone walls are a key component of these systems and are often constructed using locally sourced stones with little or no binding material.

Terrace restoration typically involves repairing or rebuilding collapsed or unstable walls, replacing displaced stones, improving structural stability, and restoring drainage functions. Soil conservation measures may include re-levelling terrace surfaces, replacing eroded soil, incorporating organic matter, and establishing vegetation cover to reduce erosion and improve slope stability. Maintaining drainage channels and preventing water accumulation behind retaining walls are also important for reducing the risk of structural damage.

The design and orientation of vineyard rows play an important role in the effectiveness of terraced systems. On sloping terrain, contour-aligned rows can reduce surface runoff, limit soil erosion, and improve water infiltration compared with downslope configurations. Modern terracing systems often combine these principles with layouts that improve accessibility and facilitate vineyard operations while maintaining environmental benefits.

Key ecosystem services

- Erosion control (*slope stabilisation*)
- Water regulation (*improved infiltration, retention, and reduced nutrient leaching*)
- Climate resilience (*enhanced stability of vineyard systems under extreme weather conditions*)
- Habitat provision and biodiversity conservation (*microhabitats for plants, invertebrates, reptiles, and small mammals*)
- Cultural heritage (*preservation of traditional knowledge and historical land-use practices*)
- Landscape aesthetics (*distinct landscape character and enhanced recreational value*)

Considerations for successful implementation

Structural maintenance and long-term planning

The long-term performance of terrace systems depends on the regular maintenance of retaining walls, embankments, and drainage infrastructure. Restoration and construction projects require careful planning and sustained investment to ensure structural stability and preserve their environmental and cultural value.

Accessibility and operational requirements

The design of terraced vineyards influences accessibility, labour requirements, and mechanisation opportunities. Modern terracing approaches increasingly seek to balance environmental objectives with operational efficiency, worker safety, and the economic viability of steep-slope viticulture.

Site-specific design and ecological integration

Successful terrace systems are adapted to local topography, soil characteristics, climatic conditions, and hydrological processes. The integration of vegetation cover, biodiversity-enhancing features, and effective water management measures strengthens erosion control, supports habitat provision, and contributes to the long-term resilience of terraced landscapes.